Privacy-Preserving Personal Identifiable Information (PII) Label Detection Using Machine Learning

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***Abstract*— In today's data-driven world, the protection of PII is of paramount importance to safeguard personal privacy. PII tags serve as crucial markers for identifying and processing sensitive information within databases. However, the authentication and registration of PII tags can be time-consuming and error prone. To address this challenge, we propose a method for privacy controlled PII tag detection that harnesses the power of machine learning (ML) combined with regular expressions. Proposed approach leverages various techniques, including feature engineering, adaptive learning, and machine learning, to extract meaningful patterns and relationships from data. By training the model on large datasets that encompass diverse PII elements such as names, addresses, phone numbers, email addresses, and social security numbers, enable it to learn and classify PII identifiers across different documents effectively. One of the key advantages of the proposed method is its ability to automate the detection of PII identifiers, thereby reducing the reliance on manual interpretation and minimizing the potential for human error. By integrating machine learning algorithms, empower organizations to efficiently identify and process sensitive information present in their databases, bolstering privacy protection measures. Moreover, this approach facilitates the development of scalable and accurate solutions for privacy based PII tag search. This advancement paves the way for enhanced data privacy across the enterprise, ensuring compliance with regulations and standards pertaining to the protection of personal information. By combining the strengths of ML and regular expressions, the proposed method enables organizations to detect and handle PII identifiers more effectively. This not only streamlines data management processes but also strengthens privacy safeguards, and more secure and privacy-aware data ecosystem.**

Key words- Privacy-Preserving, PII Label Detection, Data Privacy, Automated Detection, Machine Learning.

I. INTRODUCTION

In the digital age, where data has become a central driver of numerous aspects of our lives, safeguarding PII has emerged as a critical concern. PII encompasses any data that can be utilized to identify or trace an individual, such as their name, address, phone number, social security number, and more. Unauthorized exposure or misuse of PII can result in severe privacy breaches, identity theft, and other detrimental consequences for both individuals and organizations. Consequently, there is an escalating demand for robust mechanisms to detect and handle PII within datasets while upholding privacy preservation. The primary objective of PII label identification is to develop resilient and precise models capable of identifying and labeling PII across diverse types of data, including structured and unstructured text. The main goal is to ensure protection of sensitive information while facilitating the secure and responsible utilization of data for various applications. PII label identification leverages ML and NLP techniques to automatically identify and label PII within textual data. This process finds application in multiple domains, including healthcare, finance, and e-commerce, where protecting sensitive information is paramount and compliance with regulations such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) is mandatory. Implementing PII label identification involves a multi-faceted approach. Initially, the dataset is preprocessed to extract textual content and transform it into a suitable representation for analysis. This may involve techniques such as tokenization, stemming, and removing stop words. Subsequently, ML algorithms, such as classification or sequence labelling models, are trained on annotated data to recognize patterns and features indicative of PII. These models learn from a large volume of labelled examples, allowing them to generalize and accurately classify PII in new and unseen data. To enhance the performance and adaptability of PII label identification models, additional techniques such as feature engineering, ensemble learning, and deep learning may be employed. Feature engineering involves extracting meaningful attributes from the data that can aid in differentiating PII from non-PII content. Ensemble learning combines predictions from multiple models to improve overall accuracy and robustness. Deep learning techniques, such as recurrent neural networks (RNNs) or transformers, excel in capturing complex patterns and dependencies in text data, thereby enhancing the detection of PII. Successful implementation of PII label identification enables organizations to comply with data protection regulations, safeguard sensitive information, and foster trust with their customers. It empowers them to implement appropriate security measures, including access controls, encryption, and anonymization, to protect PII throughout its lifecycle. Moreover, by automating the identification process, the burden on manual inspection and potential human error is alleviated, leading to greater efficiency and effectiveness in PII handling.

Purpose of PII label Detection-The objective of this paper is to exhibit a novel methodology for detecting and labelling PII in a privacy-preserving manner using ML techniques. The proposed method addresses the challenge of efficiently and accurately identifying PII labels within datasets while ensuring strict privacy preservation measures. It recognizes the increasing volume and complexity of data in the digital

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age, as well as the potential risks associated with mishandling PII, such as privacy breaches and identity theft.

II. RELATED WORK AND LITERATURE SURVEY

Madelon Hulsebos *et al* [1] proposed sherlock is a multidimensional DNN for semantic type analysis. Trained Sherlock on 686,765 rows of data from the VizNet corpus, mapping 78 semantic types to rows of data from DBpedia. Described each sentence corresponding to 1,588 features describing the results of column values, character classifications, text words, and sentence vectors. Sherlock surpassed the consensus of ML libraries, dictionaries, and regular programming models to achieve a support weighted F1 score of 0.89. Dan Zhang *et al.* proposed hybrid ML model that can detect expressions in rows in a table by exploiting signals from the contents and values of rows. Sato combines deep learning models learned in large-scale collections of tables with topic models and predictive models [2]. Francesco Gargiulo *et al* proposed DNN for Hierarchical Extremely Multi-label Text Classification which provides a deep learning method for hierarchical cloud multi-label text classification. The plan is based on a hierarchical tracking mechanism that captures the relationship between letters and words in the document [3].

"DCOM: A Linear Mapper for Semantic Data Type Detection" introduces the DCOM model. The model uses the system's data and uses word embeddings to represent the content of the text. The drawback of this paper is that this article does not provide a comparison with the most advanced methods available for semantic information retrieval [4] [6]. Kavitha Srinivas *et al* proposed Semantic Feature Discovery with Code Mining and Semantic Type Detection demonstrate Semantic Feature Discovery, which enables users to interactively explore features semantically discovered from existing data science code and external knowledge [5]. Christoph Bier *et al* proposed detection and Labeling of PII in E-mails proposed an automated detection system for PII in e-mails and connected it to a usage control infrastructure [7].

Eidan. J. Rosado *et al* [8] proposed PII-CODEX: A Python library for PII Detection, Categorization, and Severity assessment python module, PII-Codex, that leverages a series of cloud service provider adapters and open-source libraries such as Microsoft Presidio and spaCy to allow for the detection and anonymization of PII but also the reasoning of why the token is classified as PII and the associated risk severity score as categorized by researchers.

Harichandana B S *et al* [9] proposed PrivPAS: A real time Privacy-Preserving AI System and applied ethics introduce PrivPAS (A real time Privacy-Preserving AI System) a novel framework to identify sensitive content. Additionally, we curate and annotate a dataset to identify and localize accessibility markers and classify whether an image is sensitive to a featured subject with a disability. Mark Gabel *et al* [10] proposed the scalable detection method for semantic clones captures semantic information from program dependency graphs (PDGs), program representations that encode information and manage dependencies between expressions and predicates. This was the first possible clone detection algorithm based on the semantic clone definition. Caitlin Brown *et al* [11] proposed Simple and Efficient Identification of PII on a public website focus on assessing the ability of the NLP tool, sentiment analysis and text classification algorithms to detect PII automatically, reliably and without too much complexity, achieved the highest accuracy at 96% and 96% F1-Score. Paulo Silva *et al* [13] proposed using NLP and ML to analyze confidential data, recommend and evaluate the use of domain recognition as a method of identifying, tracking and validating identifiable data. used best-known natural language tools (NLTK, Stanford CoreNLP and spaCy) in our experiments.

Roger Alan Stein *et al* [14] proposed employing word embeddings to analyze hierarchical text classification application of those models and algorithms on this specific problem by means of experimentation and analysis. Through testing and analysis, these models and methods are used to use word embeddings of this particular problem to define hierarchical labels.

Presently, different AI and profound learning models have been created for PII mark ID, including rule-based approaches, highlight based approaches, and brain network-based approaches. The existing approaches have shown promising outcomes in distinguishing PII data in text information. Be that as it may, there is as yet a requirement for development concerning exactness, productivity, and versatility, particularly in managing complicated and uncertain cases [12] [15] [16].

III. DESIGN AND IMPLEMENTATION

To build a PII label identification system that effectively protects organizations from data breaches and ensures compliance with regulations, it is crucial to understand the flow and steps involved in identifying personal identifiable information labels. The detailed process for implementation of proposed system as shown in figure 1.

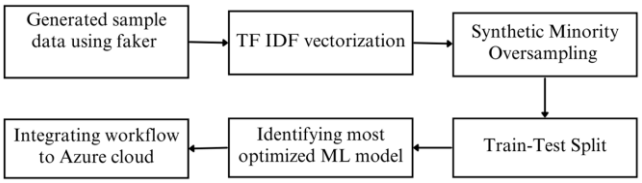


Fig.1. Flow Diagram for PII identification system

Data Collection: The initial step involves collecting sample data using the faker library in python. Faker allows generating synthetic data for various data types like names, addresses, phone numbers, etc. This sample data is generated and saved to a file or database. It serves as a representative dataset for training and testing the PII label identification system.

Data Pre-processing: Data pre-processing is a crucial stage in NLP that transforms raw text data into a format suitable for AI models. It involves several steps such as tokenization, stop word removal, stemming, lemmatization, and entity recognition. Tokenization splits the text into individual words or tokens, facilitating the extraction of meaningful insights and patterns from the data.

Text Vectorization: The next step is text vectorization, which is essential for training machine learning models. TF-IDF (Term Frequency-Inverse Document Frequency) is a popular text vectorization method used in NLP and information

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retrieval. TF-IDF converts a collection of text documents into a matrix of token (word) counts, weighted by the importance of each word in the document. The TF-IDF weight of a word in a document is calculated based on the frequency of the word in the document and its presence in other documents within the corpus. This vectorization process transforms textual data into numerical representations that can be utilized by machine learning algorithms.

Class Imbalance Handling: In PII label identification, class imbalance can occur when the number of non-PII instances significantly outweighs the PII instances. To address this issue and improve the performance of the model, the Synthetic Minority Oversampling Technique (SMOTE) can be employed. SMOTE is an oversampling technique commonly used in machine learning to address class imbalance. By generating synthetic instances of the minority class (PII instances), SMOTE helps balance the dataset, ensuring that the model receives sufficient training on PII instances for accurate detection.

ML Model Selection: Choosing the most suitable ML model is a critical step in any ML development. The performance of a model depends on factors such as data characteristics, problem complexity, and available resources. For this PII label identification system, SVM, Gaussian Naïve Bayes, and Random Forest algorithms were chosen based on their performance on specific PII label types, such as text or numerical data.

Regular Expression Matching: To further enhance the performance metrics, regular expression matching is combined with the ML models. Regular expressions provide a powerful and flexible way to define patterns and extract information from text. By incorporating regular expressions into the PII label identification system, specific patterns associated with PII, such as phone numbers or email addresses, can be accurately detected and labeled.

By following these steps, the PII label identification system can effectively process the collected data, perform data pre-processing, vectorize the text, handle class imbalance using SMOTE, select suitable ML models, and incorporate regular expression matching. This comprehensive approach enables the system to accurately identify and label PII instances within datasets, providing organizations with enhanced protection against data breaches, compliance with regulations, and improved data privacy.

IV. Simulation Results and Analysis

Proposed work generated 70,000 columns of PII data for training, validation and testing. The data was generated such that 10,000 rows of data are generated for each PII label-name, email address, address, IPv4, IPv6 address, SSN and phone number. Further splitting dataset into training and test set and used 20% data for testing and 80% data for training.

The learning curve represents the graphical depiction of how the performance of a model improves as the amount of training data increases. It shows the relationship between the training and validation loss (or accuracy) with respect to the size of the trained data. These curves demonstrate the model's ability to learn from data and its convergence as more training data is utilized. The learning curves can provide insights into potential issues such as overfitting or underfitting.

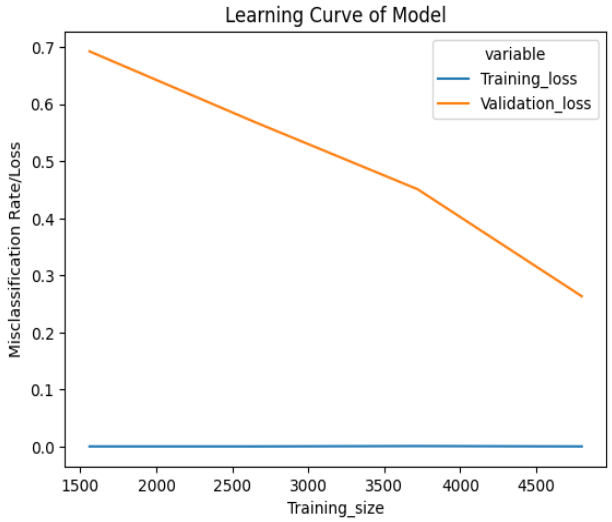


Fig.2. Training and validation loss for TF-IDF SVC algorithm

Figure 2 shows the training and validation loss for TF-IDF SVC algorithm, as the training size increases, the misclassification loss decreases. To address this, SMOTE analysis was performed to mitigate the effects of class imbalance and improve the model's ability to accurately classify PII instances. Objective is to enhance the overall performance and generalization capability of the TF-IDF SVC algorithm in identifying PII labels. The training accuracy 99.51% obtained.

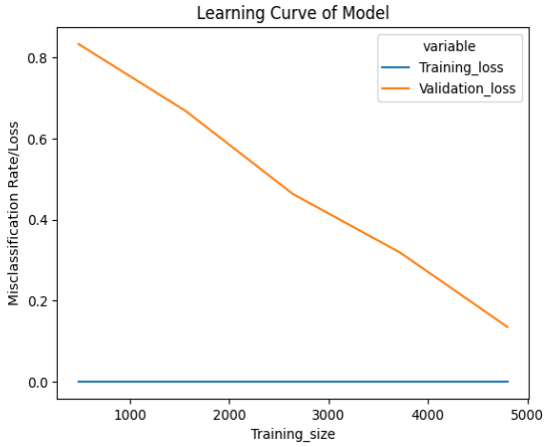


Fig.3. Training and validation loss for TF-IDF GNB algorithm

Figure 3 shows the training and validation loss for TF-IDF GNB algorithm, as the training size increases, the misclassification loss decreases. However, the application of SMOTE analysis helps counter this issue by addressing class imbalance and enhancing the model's ability to accurately classify PII instances. The objective is to improve the overall performance and generalization capability of the TF-IDF GNB algorithm in identifying PII labels. Obtained 100% accuracy. Figure 4 shows the training and validation loss for TF-IDF Random Forest algorithm, as the training size increases, the misclassification loss decreases. However, the application of SMOTE analysis helps counter this issue by addressing class imbalance and enhancing the model's ability to accurately classify PII instances. Objective is to improve the overall performance and generalization capability of the

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TF-IDF Random Forest algorithm in identifying PII labels. The training accuracy obtained 99.85%.

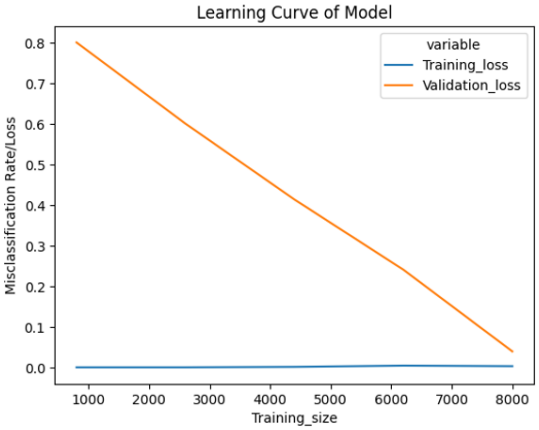


Fig.4. Training and validation loss for TF-IDF Random Forest algorithm

The comparative analysis presented in the table 1 and the learning curves help assess the performance, stability, and convergence of the different models. Based on the results, it is observed that the TF-IDF Random Forest model performs well in terms of accuracy, cross-validation score, and stability. Therefore, this model is saved as a pickle file and integrated into the final system for PII label detection.

TABLE 1. ACCURACY, CROSS-VALIDATION SCORE, STANDARD DEVIATION SCORE OF PII DETECTION

Model Used	Accuracy	Cross Validation Score	Standard Deviation
TF-IDF SVC	92.92%	73.68%	0.0111
TF-IDF-GNB	87.75%	86.56%	0.0071
TF-IDF-Random Forest	96.6%	95.91%	0.0051
Existing Solution- Sherlock	89%		

In the PII label detection system, various performance metrics were calculated to evaluate and compare the effectiveness of different models. The Table 1 indicates the comparison of different performance measures across different models.

Accuracy: Accuracy measures the proportion of correctly predicted PII labels out of the total instances. It provides an overall assessment of how well the model performs in correctly identifying PII labels. In the comparison table, the accuracy column showcases the accuracy achieved by each model. TF-IDF Random Forest model achieved an accuracy of 96.6%, indicating that it correctly predicted 96.6% of the PII labels.

Cross-validation score: Cross-validation is a technique used to estimate how well a model generalizes on unseen data. It involves partitioning the dataset into multiple subsets (folds), training the model on some folds, and evaluating its performance on the remaining folds. The cross-validation score provides an average measure of the model's performance across different folds. In the comparison table, the mean cross-validation score is presented for each model. The TF-IDF Random Forest model achieved a cross-validation score of 95.91%, indicating its strong generalization capability.

Standard deviation: Standard deviation measures the variability or spread of the cross-validation scores. A lower standard deviation suggests less variation in the performance of the model across different folds, indicating more consistent results. In the comparison table, the standard deviation of the cross-validation accuracy is presented. It is observed that the TF-IDF Random Forest model had the lowest standard deviation of 0.0051 among all the proposed models, indicating its stability and robustness.

V. CONCLUSION AND FUTURE SCOPE

In this paper privacy-preserving approach for detecting and labelling PII using ML techniques. The proposed methodology addresses the challenge of efficiently and accurately identifying PII labels within datasets while ensuring strict privacy preservation measures. It starts with collecting sample data using the faker python library, which generates fake data for various purposes. Data pre-processing techniques, including tokenization, stop word removal, stemming, lemmatization, and context awareness, are applied to transform the raw text data into a structured format suitable for ML models. Text vectorization using TF-IDF method is employed to convert the text documents into a matrix of token counts, weighted by the significance of each word in the document. Additionally, the synthetic mminority Oversampling Technique (SMOTE) is utilized to address class imbalance and enhance the detection of PII instances. Several ML models, namely Support Vector Machine (SVM), Gaussian Naïve Bayes, and Random Forest, are evaluated for PII label detection. The performance metrics, including accuracy, mean cross-validation score, and standard deviation of cross-validation accuracy, are calculated and compared to assess the effectiveness and stability of the models. Based on the obtained results, the TF-IDF Random Forest model gives highest accuracy, cross-validation score, and the least standard deviation among the proposed models. This model is selected as the best performing model and integrated into the final PII label detection system. The system effectively identifies and classifies PII, reducing the risk of data breaches and ensuring compliance with privacy regulations. This paper emphasizes the importance of continuously updating and refining the PII detection system to adapt to evolving data trends, language patterns, and privacy regulations. Regular monitoring, feedback incorporation, and compliance with privacy requirements are necessary to maintain the accuracy and effectiveness of the system. Overall, the proposed privacy preserving PII label detection approach using ML techniques provides a valuable solution for organizations to protect sensitive information, mitigate data breaches, and ensure compliance with data privacy regulations.

Future scope-While achieving a 96.6% accuracy rate in PII label identification is commendable, there is always room for enhancement and improvement. Here are some potential areas to consider for enhancing the system:

Expanded Training Data: Increasing the diversity and volume of the training data can improve the system's ability to handle various data sources, languages, and PII patterns. Incorporating more labelled examples of PII instances and non-PII instances can enhance the model's generalization capabilities.

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Fine-tuning and Model Selection: Experimenting with different ML algorithms, model architectures, and hyperparameter tuning can potentially improve the system's accuracy. Fine-tuning the existing models or exploring newer, more advanced models specifically designed for PII label identification can be beneficial.

Handling Contextual Information: Incorporating contextual information surrounding the PII instances can enhance the system's accuracy. Considering the surrounding words, sentence structure, or document context can help reduce false positives and false negatives, especially in ambiguous cases.

This work can help organizations and companies of any size to analyse and understand the PII labels and identify them effectively. The above enhancements can contribute to further improving the accuracy, efficiency, adaptability, and overall effectiveness of the PII label identification system. Regular evaluation, feedback from users, and collaboration with domain experts are crucial in identifying areas for improvement and ensuring the system stays aligned with evolving needs and challenges in the field of data privacy.

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